

Elouan Charbonnier

Malware Analyst & Reverse Engineer

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Profile

Master's graduate in Cryptology and Computer Security, specialising in malware analysis and reverse engineering, skills I developed through an internship at eShard and a series of personal projects. I am looking to apply this expertise in a role spanning both technical analysis and Cyber Threat Intelligence.

Education

University of Bordeaux <i>MSc Cryptology & Computer Security</i>	Bordeaux 2023–2025
University of Bordeaux <i>BSc Mathematics & Computer Science</i>	Bordeaux 2020–2023

Professional Experience

eShard <i>Implementation, Detection and Bypass of Anti-Emulation Techniques on Windows</i> Reverse engineering internship focused on analysing, implementing and bypassing anti-VM/anti-emulation techniques within eShard's TTD-based platform (esReverse).	Pessac, France Mar–Aug 2025
- Implemented a wide range of anti-VM techniques (CPUID, RDTSC, Windows API, ...) in x86 assembly and C - Designed an automated binary generation pipeline embedding anti-VM techniques, used to validate detection coverage - Reverse-engineered malware samples to identify anti-emulation markers (Ghidra, Binary Ninja) - Automated detection of suspicious behaviours through TTD scripting (Python) - Bypassed anti-VM protections via Windows API patching (WinDbg) and QEMU configuration tuning - Produced educational content (YouTube playlist) and a technical article published on eShard's blog - Documented all research under Jupyter Notebooks	

Projects

- GhidraMAT**: Ghidra scripting framework for automated static detection of malicious behaviours (v0.3). Features:
 - Detection of anti-VM and anti-debug techniques
 - Structured JSON/TXT report export
- Malauto**: WinDbg extension for malware analysis automation on Windows: conditional breakpoints, PE memory region dumping, structured report generation.
- Wine APT29**: Reverse engineering and behavioural analysis of the WINELOADER/GRAPELOADER malware (APT29, 2025): execution flow reconstruction, dynamic API resolution deobfuscation, persistence mechanism analysis, C2 communication identification, and evasion technique mapping.
- Panoptiv**: Windows API hooking DLL designed to bypass anti-VM protections.

Technical Skills

OS : Linux, Windows	Static analysis : Ghidra, Binary Ninja
Languages : Python, C, x86/x64 Assembly, $\LaTeX$	Dynamic analysis : WinDbg, GDB, Reven TTD
Virtualisation : QEMU, VMware, VirtualBox, FLARE-VM	CTI : MITRE ATT&CK, YARA

Publications

- 2025: Thesis — “Detection of Anti-VM Protections in Malware” [PDF]
- 2025: Article — “Anti-Emulation Techniques on Windows” — eShard [Article]
- 2025: Report — “Linux Kernel Rootkits in 2025” [PDF]

Languages

French: Native
English: B2 — Linguaskill certification

Interests

- 🧭: Hiking
- ⛺: Bivouacking